

For each prior $\pi_k, k = 1 \dots K$

Phase 1: Hamiltonian Monte Carlo (HMC)

Observations
 $\mathcal{D} = \{x_i\}_{i=1}^N$

Credal Set
 $\Pi = \{\pi_1, \dots, \pi_K\}$

Initialise
 $\theta^{(0)} = 0$

Sample momentum
 $p \sim \mathcal{N}(0, I)$

MH Accept/Reject
 $\alpha = \min\left(1, \frac{e^{-H(\tilde{\theta}, \tilde{p})}}{e^{-H(\theta, p)}}\right)$

Leapfrog integrator
 $L = 15$ steps, $\varepsilon = 0.005$
 $\tilde{\theta} = \theta + \varepsilon p$
 $\tilde{p} = p - \varepsilon \nabla U(\theta)$

reject:
keep θ

accept $\tilde{\theta}$
Warmup: 1000 steps (discard)

$$U(\theta) = -\log p(\theta|x, \pi_k) \\ = -\log \mathcal{L}(x|\theta) - \log \pi_k(\theta)$$

Posterior Cache
 $\{\theta_k^{(s)}\}_{s=1}^{3000}$ saved to disk
reused across all training runs